Sahaj Bhadja

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Summary

Software engineer with 2+ years developing, testing, debugging, and optimizing production applications and shared services. Experienced in C++, object-oriented programming, client-server systems, SQL, secure APIs, CI/CD, automated testing, performance analysis, and cross-functional delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built production APIs and Azure SQL services for 500+ users, optimizing indexed queries, schemas, and execution paths to reduce response latency from 1.2 seconds to 350 milliseconds.
- Refactored a fault-tolerant job orchestration system with explicit states and failure isolation, tripling throughput while improving shared component reuse, observability, debugging, recovery, and production reliability under load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed client-server REST services handling 10,000+ daily requests, implementing secure authentication, validation, persistence, and error handling while debugging daily production issues affecting more than 150 users.
- Created multithreaded processing for 1,000+ files with AWS RDS persistence, progress monitoring, automated validation, and failure recovery, reducing turnaround 60% under concurrent production workloads while improving reliability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python and PyTorch simulation pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, performance profiling, automated validation checks, and systematic debugging.
- Developed modular spatial-temporal components combining map geometry and agent context, improving closed-loop stability 15% while documenting designs and validating behavior across diverse simulated operating scenarios and workloads.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed object-oriented services and data integrations using AWS Glue, S3, SQL, filtering, pagination, and caching, improving production response times 30% and processing efficiency 68% across production applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating bug investigation, regression verification, and reliable deployments across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular services across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, securely processing more than two million Azure SQL records through documented interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational visibility.

Skills

Languages & Architecture: C++, Python, Java, JavaScript, TypeScript, SQL, object-oriented programming, data structures, algorithms, client-server architecture, REST APIs, microservices
Databases & Web: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, relational databases, schema design, query optimization, caching, React, Node.js, HTML, CSS, WebSockets
Testing & Development: Unit testing, integration testing, test-driven development, debugging, profiling, performance analysis, secure coding, Visual Studio familiarity, Git, Jira, PyTest, Mockito
Delivery & Platforms: Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, Linux, Windows, monitoring, logging, failure recovery, documentation, production support, Agile

Projects

CPU Scheduling Simulator | Live | *C++, React, EJ-2, Algorithms, Performance Analysis*
- Implemented six scheduling algorithms in C++ with interactive controls and Gantt visualization, validating edge cases and comparing performance metrics across more than 1,000 simulated process combinations.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built a modular Java and TypeScript database tool supporting 10+ engines with schema-aware SQL generation, query validation, metadata exploration, error handling, automated testing, and Docker-backed CI/CD deployment.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a client-server platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, WebSockets, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.